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Chapter 23

Surface Area, Volume and Capacity

Class - 8 Concise Mathematics Selina



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Exercise 23(A)

Question 1(i)

The volume of a cuboid is 4800 cm^3 . If its length is 24 cm and breadth is 20 cm; its height is :

1. 5 cm

2. 10 cm

3. 12 cm



4. 15 cm

Answer

Given:

The length of the cuboid = 24 cm.

The breadth of the cuboid = 20 cm.

Let h be the height of the cuboid.

The volume of a cuboid is 4800 cm^3 .

As we know that volume of cuboid = length $\times$ breadth $\times$ height

$$\Rightarrow 24 \times 20 \times h = 4800 \text{ cm}^3$$

$$\Rightarrow 480 \times h = 4800$$

$$\Rightarrow h = \frac{4800}{480}$$

$$\Rightarrow h = 10 \text{ cm}$$

Hence, option 2 is the correct option.

Question 1(ii)

The length, breadth and height of cuboid are in the ratio 3 : 2 : 1 and its volume is 162 cm^3 ; the longest side of the cuboid is:

1. 6 cm

2. 9 cm

3. 18 cm

4. 54 cm

Answer

Given:

The length, breadth and height of cuboid are in the ratio 3 : 2 : 1.

The volume of the cuboid = 162 cm^3 .

Let the length, breadth and height of cuboid be $3a$, $2a$ and a , respectively.

As we know that volume of cuboid = length x breadth x height

$$\Rightarrow 3a \times 2a \times a = 162 \text{ cm}^3$$

$$\Rightarrow 6a^3 = 162$$

$$\Rightarrow a^3 = \frac{162}{6}$$

$$\Rightarrow a^3 = 27$$

$$\Rightarrow a = \sqrt[3]{27}$$

$$\Rightarrow a = 3 \text{ cm}$$

Largest length of the cuboid = $3a$

$$= 3 \times 3 \text{ cm}$$

$$= 9 \text{ cm}$$

Hence, option 2 is the correct option.

Question 1(iii)

The length of a cuboid is doubled, breadth is halved and height is tripled, the volume of the cuboid will become :

1. 1.5 times
2. 2 times
3. three times
4. six times

Answer

Let l , b and h be the length, breadth and height of the cuboid.

So, new length, breadth and height are $2l$, $\frac{1}{2}b$ and $3h$.

As we know that volume of cuboid = length $\times$ breadth $\times$ height

$$= l \times b \times h$$

$$= lbh$$

$$\text{Volume of new cuboid} = 2l \times \frac{1}{2}b \times 3h$$

$$= \cancel{2}l \times \frac{1}{\cancel{2}}b \times 3h$$

$$= 3lbh$$

Hence, option 3 is the correct option.

Question 1(iv)

Each side of a cube is tripled, its surface area will become :

1. 3 times
2. six times
3. nine times
4. 27 times

Answer

Let s be the side of the cube.

So, side of new cube = $3s$

As we know that surface area of the cube = $6 \times (\text{side})^2$

$$= 6s^2$$

The surface area of new cube = $6 \times (3s)^2$

$$= 6 \times 9s^2$$

$$= 9 \times 6s^2$$

$$= 9 \times \text{Surface area of the cube}$$

Hence, option 3 is the correct option.

Question 1(v)

A cuboid has a total surface area of 80 m^2 and the lateral surface area of 50 m^2 ; the area of its base is :

1. 30 m^2

2. 60 m^2

3. 15 m^2

4. 10 m^2

Answer

The total surface area of the cuboid = 80 m^2

The lateral surface area of the cuboid = 50 m^2

As we know, the total surface area of the cuboid = $2(l \times b + b \times h + h \times l)$

$$\Rightarrow 80 = 2(l \times b + b \times h + h \times l)$$

$$\Rightarrow 80 = 2(l \times b) + 2(b \times h) + 2(h \times l) \dots\dots\dots(1)$$

The lateral surface area of the cuboid = $2(l + b)h = 2(l \times h + b \times h)$

$$\Rightarrow 50 = 2(l \times h) + 2(b \times h) \dots\dots\dots(2)$$

And, area of base = $l \times b$

Putting the value of $2(l \times h) + 2(b \times h)$ from equation (2) in equation (1), we get

$$\Rightarrow 80 = 2(l \times b) + 50$$

$$\Rightarrow 2(l \times b) = 80 - 50$$

$$\Rightarrow 2(l \times b) = 30$$

$$\Rightarrow l \times b = \frac{30}{2}$$

$$\Rightarrow l \times b = 15 \text{ m}^2$$

Hence, option 3 is the correct option.

Question 2

The length, the breadth and the height of a cuboid are in the ratio 5 : 3 : 2. If its volume is 240 cm^3 , find its dimensions. Also, find the total surface area of the cuboid.

Answer

Given:

The length, breadth and height of a cuboid are in the ratio 5 : 3 : 2.

The volume of the cuboid = 240 cm^3 .

Let the length, breadth and height of cuboid be $5a$, $3a$ and $2a$.

As we know that the volume of the cuboid = $l \times b \times h$

$$\Rightarrow 5a \times 3a \times 2a = 240 \text{ cm}^3$$

$$\Rightarrow 30a^3 = 240$$

$$\Rightarrow a^3 = \frac{240}{30}$$

$$\Rightarrow a^3 = 8$$

$$\Rightarrow a = \sqrt[3]{8}$$

$$\Rightarrow a = 2 \text{ cm}$$

Thus, the length of the cuboid = $5a = 5 \times 2 = 10 \text{ cm}$

Breadth of the cuboid = $3a = 3 \times 2 = 6 \text{ cm}$

Height of the cuboid = $2a = 2 \times 2 = 4 \text{ cm}$

As we know, the total surface area of the cuboid = $2(l \times b + b \times h + h \times l)$

$$= 2(10 \times 6 + 6 \times 4 + 4 \times 10) \text{ cm}^2$$

$$= 2(60 + 24 + 40) \text{ cm}^2$$

$$= 2 \times 124 \text{ cm}^2$$

$$= 248 \text{ cm}^2$$

Hence, the dimensions of the cuboid are 10 cm , 6 cm and 4 cm and the total surface area is 248 cm².

Question 3

The length, breadth and height of a cuboid are in the ratio 6 : 5 : 3. If its total surface area is 504 cm², find its dimensions. Also, find the volume of the cuboid.

Answer

Given:

The length, breadth and height of a cuboid are in the ratio 6 : 5 : 3.

The total surface area of the cuboid = 504 cm^2 .

Let the length, breadth and height of cuboid be $6a$, $5a$ and $3a$.

As we know, the total surface area of the cuboid = $2(l \times b + b \times h + h \times l)$

$$\Rightarrow 2(6a \times 5a + 5a \times 3a + 3a \times 6a) = 504$$

$$\Rightarrow 2(30a^2 + 15a^2 + 18a^2) = 504$$

$$\Rightarrow 2 \times 63a^2 = 504$$

$$\Rightarrow 126a^2 = 504$$

$$\Rightarrow a^2 = \frac{504}{126}$$

$$\Rightarrow a^2 = 4$$

$$\Rightarrow a = \sqrt{4}$$

$$\Rightarrow a = 2$$

Thus, length of the cuboid = $6a = 6 \times 2 \text{ cm} = 12 \text{ cm}$

Breadth of the cuboid = $5a = 5 \times 2 \text{ cm} = 10 \text{ cm}$

Height of the cuboid = $3a = 3 \times 2 \text{ cm} = 6 \text{ cm}$

As we know that volume of cuboid = $l \times b \times h$

$$= 12 \times 10 \times 6 \text{ cm}^3$$

$$= 720 \text{ cm}^3$$

Hence, the dimensions of the cuboid are 12 cm , 10 cm and 6 cm and the volume is 720 cm^3 .

Question 4

Find the length of each edge of a cube, if its volume is :

(i) 216 cm^3

(ii) 1.728 m^3

Answer

(i) **Given:**

Volume of the cube = 216 cm^3

Let s be the side of cube.

As we know that volume of cube = side^3

$$\Rightarrow s^3 = 216$$

$$\Rightarrow s = \sqrt[3]{216}$$

$$\Rightarrow s = 6 \text{ cm}$$

Hence, the length of each edge of cube is 6 cm.

(ii) **Given:**

Volume of the cube = 1.728 m^3

Let s be the side of cube.

As we know that volume of cube = side³

$$\Rightarrow s^3 = 1.728$$

$$\Rightarrow s = \sqrt[3]{1.728}$$

$$\Rightarrow s = 1.2 \text{ m}$$

Hence, the length of each edge of cube is 1.2 m.

Question 5

The total surface area of a cube is 216 cm². Find its volume.

Answer

Given:

The total surface area of a cube = 216 cm²

Let s be the side of cube.

As we know that total surface area of cube = 6 x side²

$$\Rightarrow 6 \times s^2 = 216$$

$$\Rightarrow s^2 = \frac{216}{6}$$

$$\Rightarrow s^2 = 36$$

$$\Rightarrow s = \sqrt{36}$$

$$\Rightarrow s = 6 \text{ cm}$$

The volume of cube = side³

$$= 6^3$$

$$= 216 \text{ cm}^3$$

Hence, the volume of the cube is 216 cm³.

Question 6

A wall 9 m long, 6 m high and 20 cm thick, is to be constructed using bricks of dimensions 30 cm, 15 cm and 10 cm. How many bricks will be required ?

Answer

Given:

The length of the wall = 9 m.

The height of the wall = 6 m.

The breadth of the wall = 20 cm = 0.2 m.

As we know that volume of wall = length x breadth x height

$$= 9 \times 6 \times 0.2 \text{ m}^3$$

$$= 10.8 \text{ m}^3$$

The length of the brick = 30 cm = 0.3 m.

The breadth of the brick = 15 cm = 0.15 m.

The height of the brick = 10 cm = 0.1 m.

As we know that volume of one brick = length x breadth x height

$$= 0.3 \times 0.15 \times 0.1 \text{ m}^3$$

$$= 0.0045 \text{ m}^3$$

Let a be the number of bricks.

$$\text{Number of bricks} = \frac{\text{Volume of wall}}{\text{Volume of one brick}}$$

$$\Rightarrow a = \frac{10.8}{0.0045}$$

$$\Rightarrow a = \frac{108000}{45}$$

$$\Rightarrow a = 2400$$

Hence, the number of bricks required to construct the wall is 2400.

Question 7

A solid cube of edge 14 cm is melted down and recast into smaller and equal cubes each of edge 2 cm. Find the number of smaller cubes obtained.

Answer

Side of solid cube = 14 cm

As we know, the volume of a solid cube = side^3

$$= (14)^3 \text{ cm}^3$$

$$= 2,744 \text{ cm}^3$$

Side of the smaller cube = 2 cm

Volume of the small cube = side³

$$= 2^3 \text{ cm}^3$$

$$= 8 \text{ cm}^3$$

Let a be the number of smaller cubes.

Volume of the solid cube = Volume of small cube x Number of smaller cubes

$$\Rightarrow 2,744 = 8 \times a$$

$$\Rightarrow a = \frac{2,744}{8}$$

$$\Rightarrow a = 343$$

Hence, the number of smaller cubes obtained is 343.

Question 8

A closed box is a cuboid in shape with length = 40 cm, breadth = 30 cm and height = 50 cm. It is made of thin metal sheet. Find the cost of metal sheets required to make 20 such boxes, if 1 m² of metal sheet costs ₹ 45.

Answer

Given:

The length of the closed box = 40 cm = 0.4 m.

The breadth of the closed box = 30 cm = 0.3 m.

The height of the closed box = 50 cm = 0.5 m.

As we know, the total surface area of the cuboid = $2(l \times b + b \times h + h \times l)$

$$= 2(0.4 \times 0.3 + 0.3 \times 0.5 + 0.5 \times 0.4) \text{ m}^2$$

$$= 2(0.12 + 0.15 + 0.2) \text{ m}^2$$

$$= 2 \times 0.47 \text{ m}^2$$

$$= 0.94 \text{ m}^2$$

Total surface area of 20 boxes = Surface area of 1 box $\times$ 20

$$= 0.94 \times 20 \text{ m}^2$$

$$= 18.8 \text{ m}^2$$

Cost of 1 m^2 sheet = ₹ 45

Total cost = Total surface area $\times$ Cost of 1 m^2 sheet

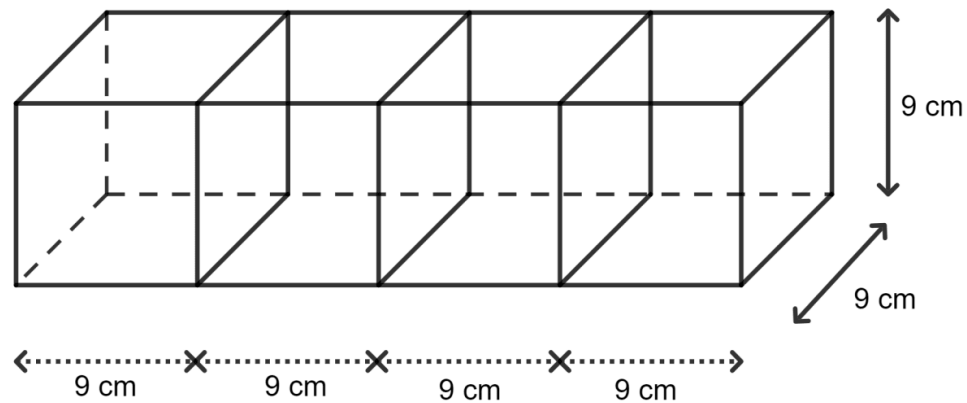
$$= ₹ 18.8 \times 45$$

$$= ₹ 846$$

Hence, the total cost of the sheet is ₹ 846.

Question 9

Four cubes, each of edge 9 cm, are joined as shown below :



Write the dimensions of the cuboid obtained. Also, find total surface area and volume.

Answer

Since the length (l) of the cuboid = $4 \times 9 \text{ cm} = 36 \text{ cm}$,

its breadth (b) = 9 cm and its height (h) = 9 cm

As we know, the total surface area of the cuboid = $2(l \times b + b \times h + h \times l)$

$$= 2(36 \times 9 + 9 \times 9 + 9 \times 36) \text{ cm}^2$$

$$= 2(324 + 81 + 324) \text{ cm}^2$$

$$= 2 \times 729 \text{ cm}^2$$

$$= 1,458 \text{ cm}^2$$

As we know, the volume of cuboid = $l \times b \times h$

$$= 36 \times 9 \times 9 \text{ cm}^3$$

$$= 2,916 \text{ cm}^3$$

Hence, the dimensions of the cuboid are 36 cm, 9cm and 9 cm, the total surface area is $1,458 \text{ cm}^2$ and the volume is $2,916 \text{ cm}^3$.

Question 10

What is the maximum length of a rod which can be kept in a rectangular box with internal dimensions 32 cm x 24 cm x 8 cm.

Answer

Given:

The length of the rectangular box = 32 cm

The breadth of the rectangular box = 24 cm

The height of the rectangular box = 8 cm

Maximum length of the rod = Length of diagonal

$$\text{Length of diagonal} = \sqrt{l^2 + b^2 + h^2}$$

$$= \sqrt{32^2 + 24^2 + 8^2}$$

$$= \sqrt{1,024 + 576 + 64}$$

$$= \sqrt{1,664}$$

$$= 8\sqrt{26}$$

$$= 40.79$$

Hence, the maximum length of a rod that can be kept in the rectangular box is 40.79 cm.

Question 11

The diagonal of a cube is $25\sqrt{3}$ m. Find its surface area.

Answer

Given:

The diagonal of a cube = $25\sqrt{3}$ m

Let s be the side of cube.

As we know that the diagonal of the cube = $\sqrt{3}$ x side

$$\Rightarrow \sqrt{3} \times s = 25\sqrt{3}$$

$$\Rightarrow \cancel{\sqrt{3}} \times s = 25 \times \cancel{\sqrt{3}}$$

$$\Rightarrow s = 25$$

As we know, the surface area of cube = $6 \times \text{side}^2$

$$= 6 \times (25)^2 \text{ m}^2$$

$$= 6 \times 625 \text{ m}^2$$

$$= 3,750 \text{ m}^2$$

Hence, the surface area of the cube is 3,750 m².

Question 12

A rectangular room is 4.5 m long, 4 m wide and 3 m high. Find the cost of white washing its walls and the roof at ₹ 15 per square metre.

Answer

Given:

The length of the rectangular room = 4.5 m

The breadth of the rectangular room = 4 m

The height of the rectangular room = 3 m

Area of white washing its walls and the roof = $l \times b + 2(l + b)h \text{ m}^2$

$$= 4.5 \times 4 + 2(4.5 + 4)3 \text{ m}^2$$

$$= 18 + 2 \times 8.5 \times 3 \text{ m}^2$$

$$= 18 + 51 \text{ m}^2$$

$$= 69 \text{ m}^2$$

Rate of white wash = ₹ 15 per square metre

Total cost of white washing = Area x Rate of white wash

$$= ₹ 69 \times 15$$

$$= ₹ 1,035$$

Hence, the cost of white washing walls and roof of rectangular room is ₹ 1,035.

Exercise 23(B)

Question 1(i)

The dimensions of a hall are 40 m x 25 m x 5 m. The number of persons which can be accommodated in the hall is (each person requires 5 m³ of air) :

1. 1000
2. 2000
3. 5000
4. 2500

Answer**Given:**

The dimensions of a hall are 40 m x 25 m x 5 m.

As we know that volume of hall = length x breadth x height

$$= 40 \times 25 \times 5 \text{ m}^3$$

$$= 5000 \text{ m}^3$$

Air required by each person = 5 m^3

Let a be the total number of persons.

Volume of the hall = number of persons $\times$ air required by each person

$$\Rightarrow 5000 = a \times 5$$

$$\Rightarrow a = \frac{5000}{5}$$

$$\Rightarrow a = 1000$$

Hence, option 1 is the correct option.

Question 1(ii)

The external dimensions of a closed rectangular box are 82 cm $\times$ 47 cm $\times$ 60 cm. If it is made of wood of 1 cm thickness, the internal dimensions of the box are :

1. 80 cm $\times$ 47 cm $\times$ 60 cm
2. 82 cm $\times$ 45 cm $\times$ 60 cm
3. 82 cm $\times$ 47 cm $\times$ 58 cm
4. 80 cm $\times$ 45 cm $\times$ 58 cm

Answer

Given:

Outer length of the box = 82 cm

Outer width of the box = 47 cm

Outer height of the box = 60 cm

Thickness of wood = 1 cm

Internal length = $82 - 1 - 1 \text{ cm} = 82 - 2 \text{ cm} = 80 \text{ cm}$

Internal width = $47 - 1 - 1 \text{ cm} = 47 - 2 \text{ cm} = 45 \text{ cm}$

Internal height = $60 - 1 - 1 \text{ cm} = 60 - 2 \text{ cm} = 58 \text{ cm}$

Hence, option 4 is the correct option.

Question 1(iii)

The outer dimensions of a closed small box are 12 cm x 12 cm x 10 cm. If it is made of 1 cm thick walls, its capacity is :

1. 1440 cm^3

2. 1000 cm^3

3. 640 cm^3

4. 800 cm^3

Answer**Given:**

Outer length of the box = 12 cm

Outer width of the box = 12 cm

Outer height of the box = 10 cm

Thickness of wood = 1 cm

Internal length = $12 - 1 - 1 \text{ cm} = 12 - 2 \text{ cm} = 10 \text{ cm}$

Internal width = $12 - 1 - 1 \text{ cm} = 12 - 2 \text{ cm} = 10 \text{ cm}$

Internal height = $10 - 1 - 1 \text{ cm} = 10 - 2 \text{ cm} = 8 \text{ cm}$

Volume of internal box = $l \times b \times h$

$$= 10 \times 10 \times 8 \text{ cm}^3$$

$$= 800 \text{ cm}^3$$

Hence, option 4 is the correct option.

Question 1(iv)

A room is 3 m long, 2 m broad and 2 m high. It has one door 2 m x 1 m; two windows each 1 m x 0.5 m; the remaining area of the walls is :

1. 16 m^2

2. 17 m^2

3. 20 m^2

4. 5 m^2

Answer**Given:**

Dimensions of the room are 3 m long, 2 m broad and 2 m high.

Dimensions of the door are 2 m x 1 m.

Dimensions of the windows are 1 m x 0.5 m.

Remaining area = Total surface area of room - (Area of door + Area of window)

Total surface area of room = $2(l + b)h$

$$= 2(3 + 2)2 \text{ m}^2$$

$$= 2 \times 5 \times 2 \text{ m}^2$$

$$= 20 \text{ m}^2$$

Area of door = $l \times b$

$$= 2 \times 1 \text{ m}^2$$

$$= 2 \text{ m}^2$$

Area of window = $l \times b$

Area of two window = $2 \times l \times b$

$$= 2 \times 1 \times 0.5 \text{ m}^2$$

$$= 1 \text{ m}^2$$

$$\text{Remaining area} = 20 - (2 + 1) \text{ m}^2$$

$$= 20 - 3 \text{ m}^2$$

$$= 17 \text{ m}^2$$

Hence, option 2 is the correct option.

Question 2

A room 5 m long, 4.5 m wide and 3.6 m high has one door 1.5 m by 2.4 m and two windows, each 1 m by 0.75 m. Find :

- (i) the area of its walls, excluding doors and windows.
- (ii) the cost of distempering its walls at the rate of ₹ 4.50 per m^2 .
- (iii) the cost of painting its roof at the rate of ₹ 9 per m^2 .

Answer

(i) **Given:**

Dimensions of the room are 5 m long, 4.5 m wide and 3.6 m high.

Dimensions of the door are 1.5 m by 2.4 m.

Dimensions of the windows are 1 m by 0.75 m.

Remaining area = Total surface area of room - (Area of door + Area of window)

Total surface area of room = $2(l + b)h$

$$= 2(5 + 4.5)3.6 \text{ m}^2$$

$$= 2 \times 9.5 \times 3.6 \text{ m}^2$$

$$= 68.4 \text{ m}^2$$

$$\text{Area of door} = l \times b$$

$$= 1.5 \times 2.4 \text{ m}^2$$

$$= 3.6 \text{ m}^2$$

$$\text{Area of window} = l \times b$$

$$\text{Area of two window} = 2 \times l \times b$$

$$= 2 \times 1 \times 0.75 \text{ m}^2$$

$$= 1.5 \text{ m}^2$$

$$\text{Remaining area} = 68.4 - (3.6 + 1.5) \text{ m}^2$$

$$= 68.4 - 5.1 \text{ m}^2$$

$$= 63.3 \text{ m}^2$$

Hence, the area of room's walls, excluding doors and windows is 63.3 m².

$$\text{(ii) The rate of distempering} = ₹ 4.50 \text{ per m}^2$$

$$\text{Total cost} = \text{Area of wall} \times \text{rate of distempering}$$

$$= ₹ 63.3 \times 4.50$$

$$= ₹ 284.85$$

Hence, the cost of distempering the walls of the room is ₹ 284.85.

(iii) The rate of painting = ₹ 9 per m^2 .

Area of roof = $l \times b$

$$= 5 \times 4.5 \text{ m}^2$$

$$= 22.5 \text{ m}^2$$

Total cost = Area of roof $\times$ Rate of painting

$$= ₹ 22.5 \times 9$$

$$= ₹ 202.5$$

Hence, the cost of painting the roof of the room is ₹ 202.5.

Question 3

The dining hall of a hotel is 75 m long, 60 m broad and 16 m high. It has five doors 4 m by 3 m each and four windows 3 m by 1.6 m each. Find the cost of :

(i) papering its walls at the rate of ₹ 12 per m^2 ;

(ii) carpeting its floor at the rate of ₹ 25 per m^2 .

Answer

(i) **Given:**

Dimensions of the room are 75 m long, 60 m broad and 16 m high.

Dimensions of the door are 4 m by 3 m.

Dimensions of the windows are 3 m by 1.6 m.

Remaining area = Total surface area of room - (Area of door + Area of window)

Total surface area of room = $2(l + b)h$

$$= 2(75 + 60)16 \text{ m}^2$$

$$= 2 \times 135 \times 16 \text{ m}^2$$

$$= 4,320 \text{ m}^2$$

Area of door = $l \times b$

Area of 5 doors = $5 \times l \times b$

$$= 5 \times 4 \times 3 \text{ m}^2$$

$$= 60 \text{ m}^2$$

Area of window = $l \times b$

Area of four window = $4 \times l \times b$

$$= 4 \times 3 \times 1.6 \text{ m}^2$$

$$= 19.2 \text{ m}^2$$

Remaining area = $4,320 - (60 + 19.2) \text{ m}^2$

$$= 4,320 - 79.2 \text{ m}^2$$

$$= 4,240.8 \text{ m}^2$$

The rate of papering = ₹ 12 per m^2

Total cost = Area of wall x rate of papering

$$= ₹ 4,240.8 \times 12$$

$$= ₹ 50,889.6$$

Hence, the cost of papering walls of dining hall is ₹ 50,889.6.

(ii) The rate of carpeting = ₹ 25 per m^2 .

Area of floor = $l \times b$

$$= 75 \times 60 \text{ m}^2$$

$$= 4500 \text{ m}^2$$

Total cost = Area of floor x Rate of carpeting

$$= ₹ 4500 \times 25$$

$$= ₹ 112,500$$

Hence, the cost of carpeting the floor of dining hall is ₹ 112,500.

Question 4

Find the volume of wood required to make a closed box of external dimensions 80 cm, 75 cm and 60 cm, the thickness of walls of the box being 2 cm throughout.

Answer

Given:

Outer length of the box = 80 cm

Outer width of the box = 75 cm

Outer height of the box = 60 cm

Volume of box = $l \times b \times h$

$$= 80 \times 75 \times 60 \text{ cm}^3$$

$$= 360,000 \text{ cm}^3$$

Thickness of wood = 2 cm

Internal length = $80 - 2 - 2 \text{ cm} = 80 - 4 \text{ cm} = 76 \text{ cm}$

Internal width = $75 - 2 - 2 \text{ cm} = 75 - 4 \text{ cm} = 71 \text{ cm}$

Internal height = $60 - 2 - 2 \text{ cm} = 60 - 4 \text{ cm} = 56 \text{ cm}$

Volume of internal box = $l \times b \times h$

$$= 76 \times 71 \times 56 \text{ cm}^3$$

$$= 302,176 \text{ cm}^3$$

Volume of wood required = $360,000 - 302,176 \text{ cm}^3$

$$= 57,824 \text{ cm}^3$$

Hence, the volume of wood required is $57,824 \text{ cm}^3$.

Question 5

A closed box measures 66 cm, 36 cm and 21 cm from outside. If its walls are made of metal sheet, 0.5 cm thick, find :

(i) the capacity of the box;

(ii) volume of metal sheet used and

(iii) weight of the box, if 1 cm³ of metal weighs 3.6 g.

Answer

(i) **Given:**

Outer length of the box = 66 cm

Outer width of the box = 36 cm

Outer height of the box = 21 cm

Volume of box = $l \times b \times h$

$$= 66 \times 36 \times 21 \text{ cm}^3$$

$$= 49,896 \text{ cm}^3$$

Thickness of metal sheet = 0.5 cm

$$\text{Internal length} = 66 - 0.5 - 0.5 \text{ cm} = 66 - 1 \text{ cm} = 65 \text{ cm}$$

$$\text{Internal width} = 36 - 0.5 - 0.5 \text{ cm} = 36 - 1 \text{ cm} = 35 \text{ cm}$$

$$\text{Internal height} = 21 - 0.5 - 0.5 \text{ cm} = 21 - 1 \text{ cm} = 20 \text{ cm}$$

Volume of internal box = $l \times b \times h$

$$= 65 \times 35 \times 20 \text{ cm}^3$$

$$= 45,500 \text{ cm}^3$$

Hence, the capacity of the box is 45,500 cm³.

(ii) Volume of metal sheet required = $49,896 - 45,500 \text{ cm}^3$

$$= 4,396 \text{ cm}^3$$

Hence, the volume of metal sheet used is 4,396 cm³.

(iii) Weight of 1 cm^3 box = 3.6 g

Weight of $4,396 \text{ cm}^3$ box = $3.6 \times 4,396 \text{ g}$

$$= 15,825.6 \text{ g}$$

Hence, the weight of the box is 15,825.6 g.

Question 6

The external length, breadth and height of a closed box are 1 m, 80 cm and 25 cm respectively. If its sides are made of 2.5 cm thick wood, find :

(i) the capacity of the box

(ii) the volume of wood used to make the box.

Answer

(i) **Given:**

Outer length of the box (L) = 1 m = 100 cm

Outer width of the box (B) = 80 cm

Outer height of the box (H) = 25 cm

Thickness of wood = 2.5 cm

Internal length (l) = 100 - 2.5 - 2.5 cm = 100 - 5 cm = 95 cm

Internal width (b) = 80 - 2.5 - 2.5 cm = 80 - 5 cm = 75 cm

Internal height (h) = 25 - 2.5 - 2.5 cm = 25 - 5 cm = 20 cm

Volume of internal box = l x b x h

$$= 95 \times 75 \times 20 \text{ cm}^3$$

$$= 142,500 \text{ cm}^3$$

Hence, the capacity of the box is 142,500 cm³.

(ii) Volume of external box = L x B x H

$$= 100 \times 80 \times 25 \text{ cm}^3$$

$$= 200,000 \text{ cm}^3$$

Volume of wood used = Volume of external box - Volume of internal box

$$200,000 - 142,500 \text{ cm}^3$$

$$= 57,500 \text{ cm}^3$$

Hence, the volume of wood used is $57,500 \text{ cm}^3$.

Question 7

Find the area of metal sheet required to make an open tank of length = 10 m, breadth = 7.5 m and depth = 3.8 m.

Answer

Given:

Length of the tank = 10 m

Breadth of the tank = 7.5 m

Height of the tank = 3.8 m

Area of metal sheet = Area of 4 walls of the tank + area of its base

$$= 2(l + b)h + l \times b$$

$$= 2(10 + 7.5)3.8 + 10 \times 7.5$$

$$= 2 \times 17.5 \times 3.8 + 75$$

$$= 133 + 75$$

$$= 208 \text{ m}^2$$

Hence, the area of metal sheet required to make an open tank is 208 m^2 .

Question 8

A tank 30 m long, 24 m wide and 4.5 m deep is to be made. It is open from the top. Find the cost of iron sheet required, at the rate of ₹ 65 per m^2 , to make the tank.

Answer

Given:

Length of the tank = 30 m

Breadth of the tank = 24 m

Height of the tank = 4.5 m

Area of 4 sides = $2(l + b)h$

$$= 2(30 + 24)4.5 \text{ m}^2$$

$$= 2 \times 54 \times 4.5 \text{ m}^2$$

$$= 486 \text{ m}^2$$

Area of floor = $l \times b$

$$= 30 \times 24 \text{ m}^2$$

$$= 720 \text{ m}^2$$

Total area = $486 \text{ m}^2 + 720 \text{ m}^2$

$$= 1,206 \text{ m}^2$$

Rate of sheet required = ₹ 65 per m^2

$$= ₹ 65 \times 1206$$

$$= ₹ 78,390$$

Hence, the total cost of iron sheet required to make tank is ₹ 78,390.

Question 9

The edges of three solid cubes are 6 cm, 8 cm and 10 cm. These cubes are melted and recasted into a single cube. Find the edge of the resulting cube.

Answer

Volume of resulting cube = Volume of three cubes.

Volume of the cube = side³

Sides of three solid cubes are 6 cm, 8 cm and 10 cm.

Let R be the side of resulting cube.

So,

$$R^3 = 6^3 + 8^3 + 10^3$$

$$\Rightarrow R^3 = 216 + 512 + 1,000$$

$$\Rightarrow R^3 = 1,728$$

$$\Rightarrow R = \sqrt[3]{1,728}$$

$$\Rightarrow R = 12 \text{ cm}$$

Hence, the edge of resulting cube is 12 cm.

Question 10

The ratio between the lengths of the edges of two cubes are in the ratio 3 : 2.

Find the ratio between their :

(i) total surface area

(ii) volume.

Answer

It is given that the ratio between the lengths of the edges of two cubes are in the ratio 3 : 2.

Let the edges of two cubes be 3a and 2a.

As we know that total surface area of cube = 6 x side²

Ratio of total surface area of two cubes = $\frac{\text{total surface area of 1st cube}}{\text{total surface area of 2nd cube}}$

$$= \frac{6 \times (3a)^2}{6 \times (2a)^2}$$

$$= \frac{6 \times 9a^2}{6 \times 4a^2}$$

$$= \frac{\cancel{6} \times 9a^2}{\cancel{6} \times 4a^2}$$

$$= \frac{9\cancel{a^2}}{4\cancel{a^2}}$$

$$= \frac{9}{4}$$

Hence, the ratio of total surface area of 2 cubs is 9 : 4.

(ii) As we know that volume of cube = side³

$$\begin{aligned}\text{Ratio of volume of two cubes} &= \frac{\text{Volume of 1st cube}}{\text{Volume of 2nd cube}} \\ &= \frac{(3a)^3}{(2a)^3} \\ &= \frac{27a^3}{8a^3} \\ &= \frac{27\cancel{a^3}}{8\cancel{a^3}} \\ &= \frac{27}{8}\end{aligned}$$

Hence, the ratio of volume of 2 cubes is 27 : 8.

Question 11

The length, breadth and height of a cuboid (rectangular solid) are 4 : 3 : 2.

(i) If its surface area is 2548 cm², find its volume.

(ii) If its volume is 3000 m³, find its surface area.

Answer

(i) It is given that the length, breadth and height of a cuboid are 4 : 3 : 2.

The surface area = 2,548 cm².

Let the length, breadth and height of a cuboid be $4a$, $3a$ and $2a$.

As we know that surface area of cuboid = $2(l \times b + b \times h + h \times l)$

$$\Rightarrow 2(4a \times 3a + 3a \times 2a + 2a \times 4a) = 2,548$$

$$\Rightarrow 2(12a^2 + 6a^2 + 8a^2) = 2,548$$

$$\Rightarrow 2 \times 26a^2 = 2,548$$

$$\Rightarrow 52a^2 = 2,548$$

$$\Rightarrow a^2 = \frac{2,548}{52}$$

$$\Rightarrow a^2 = 49$$

$$\Rightarrow a = \sqrt{49}$$

$$\Rightarrow a = 7 \text{ cm}$$

Thus, length of the cuboid = $4a = 4 \times 7 \text{ cm} = 28 \text{ cm}$

Breadth of the cuboid = $3a = 3 \times 7 \text{ cm} = 21 \text{ cm}$

Height of the cuboid = $2a = 2 \times 7 \text{ cm} = 14 \text{ cm}$

Volume of the cube = $l \times b \times h$

$$= 28 \times 21 \times 14 \text{ cm}^3$$

$$= 8,232 \text{ cm}^3$$

Hence, the volume of the cube is $8,232 \text{ cm}^3$.

(ii) The volume of the cube = $3,000 \text{ m}^3$

As we know that volume of cube = $l \times b \times h$

$$\Rightarrow 4a \times 3a \times 2a = 3,000$$

$$\Rightarrow 24a^3 = 3,000$$

$$\Rightarrow a^3 = \frac{3,000}{24}$$

$$\Rightarrow a^3 = 125$$

$$\Rightarrow a = \sqrt[3]{125}$$

$$\Rightarrow a = 5 \text{ m}$$

Thus, length of the cuboid = $4a = 4 \times 5 \text{ m} = 20 \text{ m}$

Breadth of the cuboid = $3a = 3 \times 5 \text{ m} = 15 \text{ m}$

Height of the cuboid = $2a = 2 \times 5 \text{ m} = 10 \text{ m}$

The surface area of cuboid = $2(l \times b + b \times h + h \times l)$

$$= 2(20 \times 15 + 15 \times 10 + 10 \times 20) \text{ m}^2$$

$$= 2(300 + 150 + 200) \text{ cm}^2$$

$$= 2 \times 650 \text{ cm}^2$$

$$= 1,300 \text{ m}^2$$

Hence, the surface area of the cuboid is $1,300 \text{ m}^2$.

Exercise 23(C)

Question 1(i)

The curved surface area of a cylinder, with length 10 cm and radius 7 cm, is :

1. 22 cm^2
2. 308 cm^2
3. 44 cm^2
4. 440 cm^2

Answer**Given:**

Radius of cylinder = 7 cm

Height of cylinder = 10 cm

As we know that curved surface area of cylinder = $2\pi rh$

$$= 2 \times \frac{22}{7} \times 7 \times 10$$

$$= 2 \times \frac{22}{\cancel{7}} \times \cancel{7} \times 10$$

$$= 2 \times 22 \times 10$$

$$= 440\text{cm}^2$$

Hence, option 4 is the correct option.

Question 1(ii)

Radius of the base of a solid cylinder is 1 cm and its length is 7 cm. Its volume is :

1. 11 cm^3

2. 22 cm^3

3. 33 cm^3

4. 44 cm^3

Answer

Given:

Radius of cylinder = 1 cm

Height of cylinder = 7 cm

As we know that volume of cylinder = $\pi r^2 h$

$$= \frac{22}{7} \times 1^2 \times 7$$

$$= \frac{22}{\cancel{7}} \times 1 \times \cancel{7}$$

$$= 22 \times 1$$

$$= 22\text{cm}^3$$

Hence, option 2 is the correct option.

Question 1(iii)

The radius and the length of a cylindrical rod are 5 cm and 10 cm respectively.

Its lateral surface area is :

1. 10 cm^2

2. 50 cm^2

3. 75 cm^2

4. none of these

Answer

Given:

Radius of cylinder = 5 cm

Height of cylinder = 10 cm

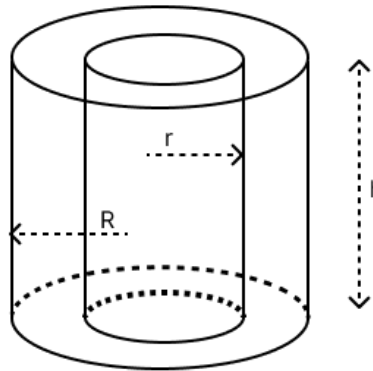
As we know that lateral surface area of cylinder = $2\pi rh$

$$= 2 \times \frac{22}{7} \times 5 \times 10$$
$$= \frac{2200}{7} \text{ cm}^2$$

Hence, option 4 is the correct option.

Question 1(iv)

The formula for the volume of the material contained in the given cylindrical pipe is :



1. $\pi(R + r) h$
2. $\pi(R - r)h$
3. $\pi(R^2 + r^2)h$
4. $\pi(R^2 - r^2)h$

Answer

Let R be the outer radius of the cylinder.

Let r be the inner radius of the cylinder.

And, h be the height of the cylinder.

As we know that volume of cylinder = $\pi r^2 h$

Volume of outer cylinder = $\pi R^2 h$

Volume of inner cylinder = $\pi r^2 h$

So, volume of the cylindrical pipe = Volume of outer cylinder - Volume of inner cylinder

$$= \pi R^2 h - \pi r^2 h$$

$$= \pi (R^2 - r^2) h$$

Hence, option 4 is the correct option.

Question 1(v)

A rectangular piece of paper 10 cm by 8 cm is rolled along its width to get the cylinder of largest size. The curved surface area of the cylinder formed is :

1. 44 cm^2

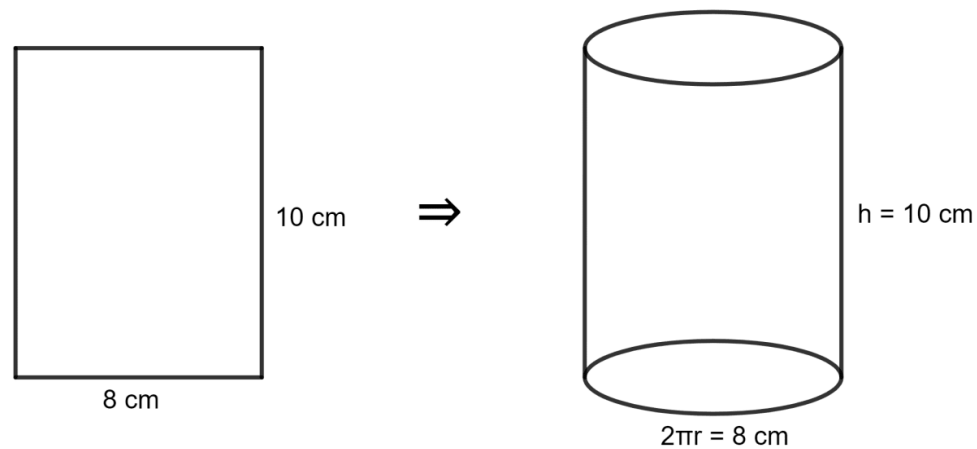
2. 20 cm^2

3. 80 cm^2

4. 80 cm^2

Answer

Clearly, width of the rectangular sheet = Circumference of the base of the cylinder



The circumference of a circle = $2\pi r$

$$\Rightarrow 2\pi r = 8$$

$$\Rightarrow 2 \times \frac{22}{7} \times r = 8$$

$$\Rightarrow \frac{44}{7} \times r = 8$$

$$\Rightarrow r = \frac{7 \times 8}{44}$$

$$\Rightarrow r = \frac{56}{44}$$

$$\Rightarrow r = \frac{14}{11}$$

Height of the cylinder formed = Length of the sheet

$$\Rightarrow h = 10 \text{ cm}$$

Curved surface area of cylinder = $2\pi rh$

$$= 2 \times \frac{22}{7} \times \frac{14}{11} \times 10 \text{ cm}^2$$

$$= \frac{44}{7} \times \frac{14}{11} \times 10 \text{ cm}^2$$

$$= \frac{616}{77} \times 10 \text{ cm}^2$$

$$= \frac{6160}{77} \text{ cm}^2$$

$$= 80 \text{ cm}^2$$

Hence, option 3 is the correct option.

Question 2

Find the height of the cylinder whose radius is 7 cm and the total surface area is 1100 cm^2 .

Answer

Given:

Radius of the cylinder = 7 cm

Total surface area of the cylinder = 1100 cm^2

Let h be the height of the cylinder.

The total surface area of cylinder = $2\pi r(r + h)$

$$\Rightarrow 2 \times \frac{22}{7} \times 7(7 + h) = 1100$$

$$\Rightarrow \frac{44}{\cancel{7}} \times \cancel{7}(7 + h) = 1100$$

$$\Rightarrow 44 \times (7 + h) = 1100$$

$$\Rightarrow 7 + h = \frac{1100}{44}$$

$$\Rightarrow 7 + h = 25$$

$$\Rightarrow h = 25 - 7$$

$$\Rightarrow h = 18$$

Hence, the height of the cylinder is 18 cm.

Question 3

The ratio between the curved surface area and the total surface area of a cylinder is 1 : 2. Find the ratio between the height and the radius of the cylinder.

Answer

It is given that the ratio between the curved surface area and the total surface area of a cylinder is 1 : 2.

$$\Rightarrow \frac{\text{Curved surface area of the cylinder}}{\text{Total surface area of the cylinder}} = \frac{1}{2}$$

$$\Rightarrow \frac{2\pi rh}{2\pi r(r + h)} = \frac{1}{2}$$

$$\Rightarrow \frac{2\pi rh}{2\pi r(r+h)} = \frac{1}{2}$$

$$\Rightarrow \frac{h}{r+h} = \frac{1}{2}$$

$$\Rightarrow 2 \times h = 1 \times (r+h)$$

$$\Rightarrow 2h = r+h$$

$$\Rightarrow 2h - h = r$$

$$\Rightarrow h = r$$

$$\Rightarrow \frac{h}{r} = \frac{1}{1}$$

Hence, the ratio between the height and radius of the cylinder is 1 : 1.

Question 4

The total surface area of a cylinder is 6512 cm^2 and the circumference of its base is 88 cm. Find :

- (i) its radius
- (ii) its volume

Answer

(i) **Given:**

The total surface area of the cylinder is 6512 cm^2 .

The circumference of its base is 88 cm

Circumference of circle = $2\pi r$

$$\Rightarrow 2 \times \frac{22}{7} \times r = 88$$

$$\Rightarrow \frac{44}{7} \times r = 88$$

$$\Rightarrow r = \frac{88 \times 7}{44}$$

$$\Rightarrow r = \frac{616}{44}$$

$$\Rightarrow r = 14$$

Hence, the radius of the cylinder is 14 cm.

(ii) Total surface area of cylinder = 6512 cm^2

$$\Rightarrow 2\pi r(r + h) = 6512 \text{ cm}^2$$

$$\Rightarrow 2 \times \frac{22}{7} \times 14 \times (14 + h) = 6512$$

$$\Rightarrow \frac{44}{7} \times 14 \times (14 + h) = 6512$$

$$\Rightarrow \frac{616}{7} \times (14 + h) = 6512$$

$$\Rightarrow 88 \times (14 + h) = 6512$$

$$\Rightarrow (14 + h) = \frac{6512}{88}$$

$$\Rightarrow (14 + h) = 74$$

$$\Rightarrow h = 74 - 14$$

$$\Rightarrow h = 60$$

Height of the cylinder = 60 cm

Volume of the cylinder = $\pi r^2 h$

$$= \frac{22}{7} \times 14^2 \times 60$$

$$= \frac{22}{7} \times 196 \times 60$$

$$= \frac{258,720}{7}$$

$$= 36,960$$

Hence, the volume of the cylinder is 36,960 cm³.

Question 5

The sum of the radius and the height of a cylinder is 37 cm and the total surface area of the cylinder is 1628 cm². Find the height and the volume of the cylinder.

Answer

Given:

The sum of the radius and the height of a cylinder is 37 cm.

Total surface area of the cylinder = 1628 cm^2

Let r be the radius and h be the height of cylinder.

So, $r + h = 37$

As we know that total surface area of cylinder = $2\pi r(r + h)$

$$\Rightarrow 2 \times \frac{22}{7} \times r \times 37 = 1628$$

$$\Rightarrow \frac{1628}{7} \times r = 1628$$

$$\Rightarrow r = \frac{7 \times 1628}{1628}$$

$$\Rightarrow r = \frac{7 \times \cancel{1628}}{\cancel{1628}}$$

$$\Rightarrow r = 7$$

It is given that $r + h = 37$

$$\Rightarrow 7 + h = 37$$

$$\Rightarrow h = 37 - 7$$

$$\Rightarrow h = 30$$

Height of the cylinder is 30 cm.

Volume of the cylinder = $\pi r^2 h$

$$= \frac{22}{7} \times 7^2 \times 30$$

$$= \frac{22}{7} \times 49 \times 30$$

$$= \frac{32,340}{7}$$

$$= 4,620$$

Hence, the height of the cylinder is 30 cm and the volume is 4,620 cm³.

Question 6

A cylindrical pillar has radius 21 cm and height 4 m. Find :

- (i) the curved surface area of the pillar
- (ii) the cost of polishing 36 such cylindrical pillars at the rate of ₹ 12 per m².

Answer

(i) **Given:**

Radius of cylindrical pillar = 21 cm

Height = 4 m = 400 cm

As we know that curved surface area of the cylinder = $2\pi rh$

$$= 2 \times \frac{22}{7} \times 21 \times 400$$

$$= \frac{369,600}{7}$$

$$= 52,800$$

Hence, the curved surface area of the cylindrical pillar is $52,800 \text{ cm}^2 = 5.28 \text{ m}^2$.

(ii) Curved surface area of 1 pillar = 5.28 m^2

Curved surface area of 36 pillar = $5.28 \text{ m}^2 \times 36$

$$= 190.08$$

Rate of polishing = ₹ 12 per m^2

Total cost = Curved surface area of 36 pillar $\times$ Rate of polishing

$$= 190.08 \times ₹ 12$$

$$= ₹ 2,280.96$$

Hence, the total cost of polishing 36 cylindrical pillars is ₹ 2,280.96.

Question 7

If the radii of two cylinders are in the ratio 4 : 3 and their heights are in the ratio 5 : 6, find the ratio of their curved surfaces.

Answer

Given:

The radii of two cylinders are in the ratio 4 : 3.

The heights of two cylinders are in the ratio 5 : 6.

The curved surface area of the cylinder = $2\pi rh$

Ratio of curved surface area of cylinders =

$$\frac{\text{Curved surface area of the first cylinder}}{\text{Curved surface area of the second cylinder}}$$

$$= \frac{2\pi r_1 h_1}{2\pi r_2 h_2}$$

$$= \frac{2\pi \times 4 \times 5}{2\pi \times 3 \times 6}$$

$$= \frac{20}{18}$$

$$= \frac{10}{9}$$

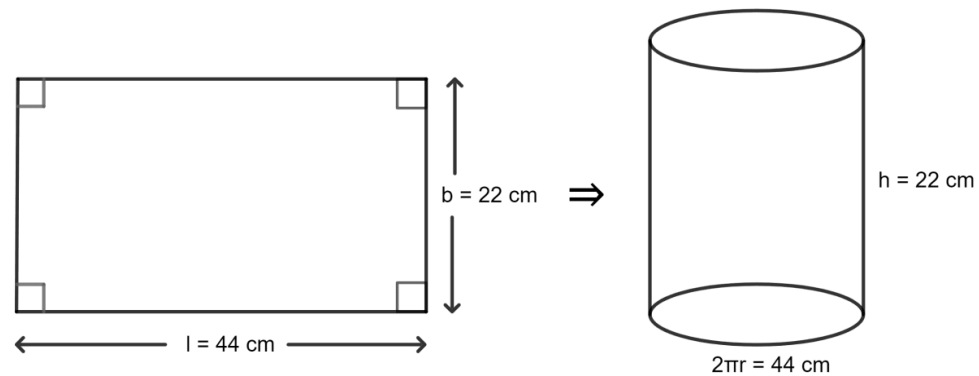
Hence, the ratio of the curved surface areas of two cylinders is 10 : 9.

Question 8

A thin rectangular card board has dimensions 44 cm x 22 cm. It is rolled along its length to get a hollow cylinder of largest size. Find the volume of the cylinder formed.

Answer

Clearly, length of the rectangular sheet = Circumference of the base of the cylinder formed



$$\Rightarrow 2\pi r = 44$$

$$\Rightarrow 2 \times \frac{22}{7} \times r = 44$$

$$\Rightarrow \frac{44}{7} \times r = 44$$

$$\Rightarrow r = \frac{7 \times 44}{44}$$

$$\Rightarrow r = \frac{7 \times \cancel{44}}{\cancel{44}}$$

$$\Rightarrow r = 7 \text{ cm}$$

Also, height of the cylinder formed = Breadth of the sheet

$$\Rightarrow h = 22 \text{ cm}$$

$$\text{Volume of the cylinder formed} = \pi r^2 h$$

$$\begin{aligned}
 &= \frac{22}{7} \times 7^2 \times 22 \\
 &= \frac{22}{\cancel{7}} \times \cancel{7} \times 7 \times 22 \\
 &= 22 \times 7 \times 22 \\
 &= 3,388 \text{ cm}^3
 \end{aligned}$$

Hence, the volume of the cylinder is 3,388 cm³.

Question 9

The ratio between the curved surface area and the total surface area of a right circular cylinder is 3 : 5. Find the ratio between the height and the radius of the cylinder.

Answer

It is given that the ratio between the curved surface area and the total surface area of a right circular cylinder is 3 : 5.

$$\begin{aligned}
 \Rightarrow \frac{\text{Curved surface area of cylinder}}{\text{Total surface area of cylinder}} &= \frac{3}{5} \\
 \Rightarrow \frac{2\pi rh}{2\pi r(r+h)} &= \frac{3}{5} \\
 \Rightarrow \frac{\cancel{2\pi} r h}{\cancel{2\pi} r(r+h)} &= \frac{3}{5} \\
 \Rightarrow \frac{h}{r+h} &= \frac{3}{5}
 \end{aligned}$$

$$\Rightarrow 5 \times h = 3 \times (r + h)$$

$$\Rightarrow 5h = 3r + 3h$$

$$\Rightarrow 5h - 3h = 3r$$

$$\Rightarrow 2h = 3r$$

$$\Rightarrow \frac{h}{r} = \frac{3}{2}$$

Hence, the ratio between the height and the radius of the cylinder is 3 : 2.

Question 10

If radii of two circular cylinders are in the ratio 3 : 4 and their heights are in the ratio 6 : 5, find the ratio of their curved surface areas.

Answer

Given:

The radii of two circular cylinders are in the ratio 3 : 4.

The heights of two circular cylinders are in the ratio 6 : 5.

Ratio of curved surface areas of cylinders =

$$\frac{\text{Curved surface area of first cylinder}}{\text{Curved surface area of second cylinder}}$$

$$= \frac{2\pi r_1 h_1}{2\pi r_2 h_2}$$

$$= \frac{\cancel{2\pi} \times 3 \times 6}{\cancel{2\pi} \times 4 \times 5}$$

$$= \frac{18}{20}$$
$$= \frac{9}{10}$$

Hence, the ratio of the curved surface areas of two cylinders is 9 : 10.

Test Yourself

Question 1(i)

The lateral surface area of a cube is 100 cm^2 ; then its volume is :

1. 25 cm^3
2. 125 cm^3
3. 625 cm^3
4. none of these

Answer

Given:

Lateral surface area of cube is 100 cm^2 .

Let s be the side of cube.

As we know lateral surface area of cube = $4 \times (\text{side})^2$

$$\Rightarrow 4 \times s^2 = 100 \text{ cm}^2$$

$$\Rightarrow s^2 = \frac{100}{4} \text{ cm}^2$$

$$\Rightarrow s^2 = 25 \text{ cm}^2$$

$$\Rightarrow s = \sqrt{25} \text{ cm}$$

$$\Rightarrow s = 5 \text{ cm}$$

Volume of the cube = $(\text{side})^3$

$$= 5^3 \text{ cm}^3$$

$$= 125 \text{ cm}^3$$

Hence, option 2 is the correct option.

Question 1(ii)

The length, the breadth and the height of a cuboid are doubled, the ratio between the volumes of the new cuboid and the original cuboid is :

$$1. 4 : 1$$

2. 1 : 4

3. 8 : 1

4. 1 : 8

Answer

Let l be the length, b be the breadth and h be the height of the original cuboid.

It is given that the length, breadth and height are doubled.

As we know, the volume of cuboid = length $\times$ breadth $\times$ height

The Ratio of the volume of the new cuboid to the original cuboid is:

$$= \frac{\text{Volume of new cuboid}}{\text{Volume of old cuboid}}$$

$$= \frac{2l \times 2b \times 2h}{l \times b \times h}$$

$$= \frac{8lbh}{lbh}$$

$$= \frac{8\cancel{lbh}}{\cancel{lbh}}$$

$$= \frac{8}{1}$$

Hence, option 3 is the correct option.

Question 1(iii)

The radius of a cylinder is doubled and its height is halved; then the new volume is :

1. same
2. 2 times
3. 4 times
4. 8 times

Answer

Let r be the radius and h be the height of original cylinder.

It is given that radius of cylinder is doubled and its height is halved.

Thus, radius of new cylinder = $2r$ and height = $\frac{1}{2}h$

As we know, the volume of cylinder = $\pi r^2 h$

Volume of original cylinder = $\frac{22}{7} r^2 h$

Volume of new cylinder:

$$= \frac{22}{7} \times (2r)^2 \times \frac{1}{2}h$$

$$= \frac{22}{7} \times 4r^2 \times \frac{1}{2}h$$

$$= \frac{4}{2} \times \frac{22}{7} r^2 h$$
$$= 2 \times \frac{22}{7} r^2 h$$

The new volume is 2 times the original volume.

Hence, option 2 is the correct option.

Question 1(iv)

The dimensions of a solid metallic cuboid are 9 cm x 8 cm x 3 cm. It is melted and recast into a cube. The edge of the cube so formed is :

1. 6 cm
2. 12 cm
3. 18 cm
4. 24 cm

Answer

It is given that the dimensions of a solid metallic cuboid are 9 cm x 8 cm x 3 cm.

The metallic cuboid is melted and recast into a cube i.e., the volume of both shapes remains the same.

Let s be the side of the cube.

Volume of cuboid = Volume of cube

$$\Rightarrow l \times b \times h = \text{side}^3$$

$$\Rightarrow s^3 = 9 \times 8 \times 3 \text{ cm}^3$$

$$\Rightarrow s^3 = 216 \text{ cm}^3$$

$$\Rightarrow s = \sqrt[3]{216} \text{ cm}$$

$$\Rightarrow s = 6 \text{ cm}$$

The edge of the cube formed is 6 cm.

Hence, option 1 is the correct option.

Question 1(v)

The volume of a cuboid is 448 cm^3 . Its height is 7 cm and the base is square.

Then each side of the base is :

1. 64 cm

2. 16 cm

3. 4 cm

4. 8 cm

Answer

Given:

Volume of the cuboid = 448 cm^3

Height of the cuboid = 7 cm

The base is square, meaning length = breadth.

Let a be the length of the cuboid.

As we know, the volume of cuboid = $l \times b \times h$.

Since the base is square, we can write:

$$\Rightarrow a \times a \times 7 = 448$$

$$\Rightarrow a^2 \times 7 = 448$$

$$\Rightarrow a^2 = \frac{448}{7}$$

$$\Rightarrow a^2 = 64$$

$$\Rightarrow a = \sqrt{64}$$

$$\Rightarrow a = 8 \text{ cm}$$

The each side of the base is 8 cm.

Hence, option 4 is the correct option.

Question 1(vi)

Statement 1: The volume of a right circular cylinder = Area of cross section (area of circle) $\times$ Distance between two circular parallel ends.

Statement 2: Lateral surface area of a closed cylinder is $2\pi r(r + h)$ cubic units.

Which of the following options is correct?

1. Both the statements are true.
2. Both the statements are false.

3. Statement 1 is true, and statement 2 is false.

4. Statement 1 is false, and statement 2 is true.

Answer

The volume of a right circular cylinder = Area of cross section (area of circle) x
Distance between two circular parallel ends.

Area of cross section = πr^2

Distance between two circular parallel ends = h

Volume = $\pi r^2 h$

So, statement 1 is true.

The lateral surface area of a cylinder is the area of its curved surface, excluding
the top and bottom bases.

Lateral surface area = $2\pi rh$ square units.

Total surface area = $2\pi r(r + h)$ square units.

So, statement 2 is false.

$\therefore$ Statement 1 is true, and statement 2 is false.

Hence, option 3 is the correct option.

Question 1(vii)

Assertion (A) : The length, breadth and height of an open cuboid are 10 cm, 12 cm and 6 cm respectively. If the thickness is 1 cm, then internal dimensions are

8 cm, 10 cm and 5 cm respectively.

Reason (R) : If l , b and h are the external dimension of an open cuboid of thickness ' x ', then its internal dimension are $(l - 2x)$, $(b - 2x)$ and $(h - x)$ respectively.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

l , b and h represent the external length, breadth, and height, respectively.

The wall has thickness ' x '.

The internal dimensions are reduced by the thickness of the walls.

The internal length = $(l - 2x)$, accounting for the thickness on both sides.

The internal breadth = $(b - 2x)$, accounting for the thickness on both sides.

The internal height = $(h - x)$, as the thickness affects only the bottom and sides, not the open top.

So, reason (R) is true.

10 cm, 12 cm and 6 cm are the external length, breadth, and height, respectively.

The wall has thickness 1 cm.

The internal length = $(10 - 2 \times 1) = (10 - 2) = 8$ cm.

The internal breadth = $(12 - 2 \times 1) = (12 - 2) = 10$ cm.

The internal height = $(6 - 1) = 5$ cm.

So, assertion (A) is true.

$\therefore$ Both A and R are correct, and R is the correct explanation for A.

Hence, option 1 is the correct option.

Question 1(viii)

Assertion (A) : Three solid silver cubes of side 6 cm, 8 cm and 10 cm are melted and recasted into a single solid cube. The side of the new cube = 2 times the side of the smallest cube.

Reason (R) : Volume of cuboid = $(l \times b \times h)$ cubic units.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

When three solid cubes with edges 6 cm, 8 cm, and 10 cm are melted and recast into a single cube, the total volume is the sum of the individual volumes:

Volume of the 6 cm cube : $6^3 = 216$ cm³

Volume of the 8 cm cube : $8^3 = 512 \text{ cm}^3$

Volume of the 10 cm cube : $10^3 = 1000 \text{ cm}^3$

Total volume = $216 + 512 + 1000 = 1728 \text{ cm}^3$

Let x cm be the side of new cube formed.

$$\Rightarrow x^3 = 1728 \text{ cm}^3$$

$$\Rightarrow x = \sqrt[3]{1728} \text{ cm}$$

$$\Rightarrow x = 12 \text{ cm}$$

$$\Rightarrow x = 2 \times 6 \text{ cm}$$

$$\Rightarrow x = 2 \times \text{side of the smallest cube melted}$$

So, assertion (A) is true.

When l, b and h represent the length, breadth, and height, respectively of the cuboid.

Volume of cuboid = $(l \times b \times h)$ cubic units.

So, reason (R) is true.

$\therefore$ Both A and R are correct, and R is not the correct explanation for A.

Hence, option 2 is the correct option.

Question 1(ix)

Assertion (A) : The length, breadth and height of the cuboid are 15 cm, 12 cm and 9 cm respectively. Lateral surface area of the cuboid = 846 cm^2 .

Reason (R) : Lateral surface area of cuboid = $2 \times h \times (l + b)$ square units.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

When l, b and h represent the length, breadth, and height, respectively of the cuboid.

By formula,

$$\text{Lateral surface area} = 2(l + b)h$$

So, reason (R) is true.

Substituting values, we get :

$$\text{Lateral surface area} = 2 \times (15 + 12) \times 9$$

$$= 2 \times 27 \times 9$$

$$= 486 \text{ cm}^2.$$

So, assertion (A) is false.

$\therefore$ A is false, but R is true.

Hence, option 4 is the correct option.

Question 1(x)

Assertion (A) : If radius of a right circular cylinder is double and the height is reduced to $\frac{1}{2}$ of the original, the ratio of volume of new cylinder thus formed to the volume of the original cylinder is 1 : 1.

Reason (R) : Volume of a cylinder = $\pi r^2 h$ where r is the radius of the circular base and h is height.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are incorrect.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

Let originally r be the radius of the circular base and h be height of the circular cylinder, then

Volume of a cylinder (v) = $\pi r^2 h$.

So, reason (R) is true.

Given,

Radius of new cylinder (R) = 2r

Height of new cylinder (H) = $\frac{1}{2} h$

Ratio of volume of new cylinder thus formed to the volume of the original cylinder,

$$\begin{aligned}\Rightarrow \frac{V}{v} &= \frac{\pi R^2 H}{\pi r^2 h} \\&= \frac{\pi \times (2r)^2 \times \frac{1}{2}h}{\pi r^2 h} \\&= \frac{\pi \times 4r^2 \times \frac{1}{2}h}{\pi r^2 h} \\&= \frac{2\pi r^2 h}{\pi r^2 h} \\&= \frac{2}{1}.\end{aligned}$$

So, assertion (A) is false.

$\therefore$ A is false, but R is true.

Hence, option 4 is the correct option.

Question 2

A cuboid is 8 m long, 12 m broad and 3.5 m high. Find its

(i) total surface area

(ii) lateral surface area

Answer

(i) **Given:**

Length of cuboid = 8 m

Breadth of cuboid = 12 m

Height of cuboid = 3.5 m

The total surface area of the cuboid = $2(l \times b + b \times h + h \times l)$

$$= 2 (8 \times 12 + 12 \times 3.5 + 3.5 \times 8) \text{ m}^2$$

$$= 2 (96 + 42 + 28) \text{ m}^2$$

$$= 2 \times 166 \text{ m}^2$$

$$= 332 \text{ m}^2$$

Hence, the total surface area of the cuboid is 332 m^2 .

(ii) Lateral surface area of cuboid = $2(l + b)h$

$$= 2(8 + 12)3.5 \text{ m}^2$$

$$= 2 \times 20 \times 3.5 \text{ m}^2$$

$$= 140 \text{ m}^2$$

Hence, the lateral surface area of the cuboid is 140 m^2 .

Question 3

How many bricks will be required for constructing a wall which is 16 m long, 3 m high and 22.5 cm thick, if each brick measures 25 cm x 11.25 cm x 6 cm ?

Answer

Given:

Length of the wall = 16 m

Breadth of the wall = 3 m

Height of the wall = 22.5 cm = 0.225 m

The volume of wall = length x breadth x height

$$= 16 \times 3 \times 0.225 \text{ m}^3$$

$$= 10.8 \text{ m}^3$$

Length of brick = 25 cm = 0.25 m.

Breadth of brick = 11.25 cm = 0.1125 m.

Height of brick = 6 cm = 0.06 m.

The volume of one brick = length x breadth x height

$$= 0.25 \times 0.1125 \times 0.06 \text{ m}^3$$

$$= 0.0016875 \text{ m}^3$$

Let a be the number of bricks.

$$\text{Number of bricks} = \frac{\text{Volume of wall}}{\text{Volume of one brick}}$$

$$\Rightarrow a = \frac{10.8}{0.0016875}$$
$$\Rightarrow a = \frac{108,000,000}{16,875}$$

$$\Rightarrow a = 6,400$$

Hence, 6,400 bricks are required to construct the wall.

Question 4

The length, breadth and height of a cuboid are in the ratio 6 : 5 : 3. If its total surface area is 504 cm^2 , find its volume.

Answer

Given:

The length, breadth and height of a cuboid are in the ratio 6 : 5 : 3.

Total surface area of cuboid = 504 cm^2 .

Let the length, breadth and height of cuboid be $6a$, $5a$ and $3a$.

As we know total surface area of cuboid = $2(l \times b + b \times h + h \times l)$

$$\Rightarrow 2(6a \times 5a + 5a \times 3a + 3a \times 6a) = 504$$

$$\Rightarrow 2(30a^2 + 15a^2 + 18a^2) = 504$$

$$\Rightarrow 2 \times 63a^2 = 504$$

$$\Rightarrow 126a^2 = 504$$

$$\Rightarrow a^2 = \frac{504}{126}$$

$$\Rightarrow a^2 = 4$$

$$\Rightarrow a = \sqrt{4}$$

$$\Rightarrow a = 2$$

Thus, length of cuboid = $6a = 6 \times 2 \text{ cm} = 12 \text{ cm}$

Breadth of cuboid = $5a = 5 \times 2 \text{ cm} = 10 \text{ cm}$

Height of cuboid = $3a = 3 \times 2 \text{ cm} = 6 \text{ cm}$

As we know that volume of cuboid = $l \times b \times h$

$$= 12 \times 10 \times 6 \text{ cm}^3$$

$$= 720 \text{ cm}^3$$

Hence, the volume of cuboid is 720 cm^3 .

Question 5

The external dimensions of an open wooden box are 65 cm, 34 cm and 25 cm.

If the box is made up of wood 2 cm thick, find the capacity of the box and the volume of wood used to make it.

Answer

Given:

Outer length of box = 65 cm

Outer width of box = 34 cm

Outer height of box = 25 cm

Volume of outer box = $l \times b \times h$

$$= 65 \times 34 \times 25 \text{ cm}^3$$

$$= 55,250 \text{ cm}^3$$

Thickness of wood = 2 cm

Internal length of box = $65 - 2 - 2 \text{ cm} = 65 - 4 \text{ cm} = 61 \text{ cm}$

Internal width of box = $34 - 2 - 2 \text{ cm} = 34 - 4 \text{ cm} = 30 \text{ cm}$

Internal height of box = $25 - 2 \text{ cm} = 23 \text{ cm}$

Volume of internal box = $l \times b \times h$

$$= 61 \times 30 \times 23 \text{ cm}^3$$

$$= 42,090 \text{ cm}^3$$

Volume of wood = Volume of outer box - Volume of internal box

$$= 55,250 - 42,090 \text{ cm}^3$$

$$= 13,160 \text{ cm}^3$$

Hence, the capacity of the box is $42,090 \text{ cm}^3$ and the volume of wood used is $13,160 \text{ cm}^3$.

Question 6

The curved surface area and the volume of a toy, cylindrical in shape, are 132 cm^2 and 462 cm^3 respectively. Find its diameter and its length.

Answer

Given:

Curved surface area of cylindrical toy = 132 cm^2

Volume of cylindrical toy = 462 cm^3

As we know that curved surface area cylinder = $2\pi rh$

$$\Rightarrow 2\pi rh = 132$$

$$\Rightarrow \pi rh = \frac{132}{2}$$

$$\Rightarrow \pi rh = 66 \dots\dots\dots(1)$$

And, volume of cylinder = $\pi r^2 h$

$$\Rightarrow \pi r^2 h = 462$$

$$\Rightarrow \pi rh \times r = 462$$

Putting the value of ' πrh ' from equation (1), we get

$$\Rightarrow 66 \times r = 462$$

$$\Rightarrow r = \frac{462}{66}$$

$$\Rightarrow r = 7 \text{ cm}$$

$$\text{Diameter} = 2 \times r$$

$$= 2 \times 7 \text{ cm} = 14 \text{ cm}$$

Putting the value of r in equation (1),

$$\Rightarrow \frac{22}{7} \times 7 \times h = 66$$

$$\Rightarrow \frac{22}{\cancel{7}} \times \cancel{7} \times h = 66$$

$$\Rightarrow 22 \times h = 66$$

$$\Rightarrow h = \frac{66}{22}$$

$$\Rightarrow h = 3$$

Hence, the diameter of cylinder is 14 cm and the length is 3 cm.

Question 7

The floor of a rectangular hall has a perimeter 250 m. If the cost of painting the four walls at the rate of ₹ 10 per m^2 is ₹ 15,000, find the height of the hall.

Answer

Given:

Perimeter of the floor of the rectangular hall = 250 m

Total cost of painting the four walls = ₹ 15,000

Rate of painting = ₹ 10 per m^2

Let l be the length, b be the breadth and h be the height of rectangular hall.

$$\text{Perimeter of floor} = 2(l + b)$$

$$\Rightarrow 2(l + b) = 250$$

$$\Rightarrow l + b = \frac{250}{2}$$

$$\Rightarrow l + b = 125 \dots\dots\dots(1)$$

The total cost of painting the four walls equals to the curved surface area of the rectangular hall.

$$\text{Total cost} = \text{Curved surface area of hall} \times \text{Rate of painting}$$

$$\Rightarrow 15,000 = \text{Curved surface area} \times 10$$

$$\Rightarrow \text{Curved surface area} = \frac{15,000}{10}$$

$$\Rightarrow \text{Curved surface area} = 1,500 \text{ m}^2$$

$$\text{Curved surface area of hall} = 2(l + b)h$$

$$\Rightarrow 1,500 = 2(l + b)h$$

Using equation (1),

$$\Rightarrow 1,500 = 2 \times 125 \times h$$

$$\Rightarrow 1,500 = 250 \times h$$

$$\Rightarrow h = \frac{1500}{250}$$

$$\Rightarrow h = 6$$

Hence, the height of the rectangular hall is 6 m.

Question 8

The length of a hall is double its breadth. Its height is 3 m. The area of its four walls (including doors and windows) is 108 m^2 , find its volume.

Answer

Given:

Let b be the breadth of the hall.

The length l of the hall is double its breadth, so $l = 2b$

The height h of the hall is 3 m.

The area of the four walls is 108 m^2 .

The area of the four walls of a hall $= 2(l + b)h$

$$\Rightarrow 2(l + b)h = 108$$

$$\Rightarrow 2(2b + b) \times 3 = 108$$

$$\Rightarrow 2 \times 3b \times 3 = 108$$

$$\Rightarrow 18b = 108$$

$$\Rightarrow b = \frac{108}{18}$$

$$\Rightarrow b = 6 \text{ m}$$

$$\text{So, } l = 2b = 2 \times 6 \text{ m} = 12 \text{ m}$$

And, volume of hall = $l \times b \times h$

$$= 12 \times 6 \times 3 \text{ m}^3$$

$$= 216 \text{ m}^3$$

Hence, the volume of the hall is 216 m^3 .

Question 9

A solid cube of side 12 cm is cut into 8 identical cubes. What will be the side of the new cube? Also, find the ratio between the surface area of the original cube and the total surface area of all the small cubes formed.

Answer

Side of the original cube = 12 cm

Let s be the side of each smaller cube.

As we know, the volume of a cube = side^3

Volume of the original cube = Volume of 8 identical cubes

$$\Rightarrow (12)^3 = 8 \times s^3$$

$$\Rightarrow 1,728 = 8 \times s^3$$

$$\Rightarrow s^3 = \frac{1,728}{8}$$

$$\Rightarrow s^3 = 216$$

$$\Rightarrow s = \sqrt[3]{216}$$

$$\Rightarrow s = 6 \text{ cm}$$

The surface area of a cube = $6 \times \text{side}^2$

Ratio of the total surface area of the original cube to the total surface area of all the smaller cubes =

$$= \frac{6 \times (12)^2}{8 \times 6 \times (6)^2}$$

$$= \frac{6 \times 144}{8 \times 216}$$

$$= \frac{864}{1728}$$

$$= \frac{1}{2}$$

Hence, the side of each smaller cube is 6 cm and the ratio of the total surface area of the original cube to that of all the smaller cubes is 1 : 2.

Question 10

The diameter of a garden roller is 1.4 m and it is 2 m long. Find the maximum area covered by it in 50 revolutions ?

Answer

Given:

Diameter of the roller = 1.4 m

Length of the roller = 2 m

Number of Revolutions = 50

$$\text{Radius of the roller} = \frac{1.4}{2} = 0.7 \text{ m}$$

As we know, the curved surface area of cylinder = $2\pi rh$

$$= 2 \times \frac{22}{7} \times 0.7 \times 2$$

$$= \frac{61.6}{7}$$

$$= 8.8$$

The curved surface area of cylinder is 8.8 m^2 .

Maximum area covered by the roller = Curved surface area x Number of revolutions

$$= 8.8 \times 50$$

$$= 440 \text{ m}^2$$

Hence, the maximum area covered by the garden roller in 50 revolutions is 440 m^2 .

Question 11

In a building, there are 24 cylindrical pillars. For each pillar, radius is 28 cm and height is 4 m. Find the total cost of painting the curved surface area of the pillars at the rate of ₹ 8 per m^2 .

Answer

Given:

Number of cylindrical pillars = 24

Radius of each pillar = 28 cm = 0.28 m

Height of each pillar = 4 m

As we know, the curved surface area of cylinder = $2\pi rh$

Curved surface area of 1 pillar =

$$= 2 \times \frac{22}{7} \times 0.28 \times 4$$

$$= \frac{49.28}{7}$$

$$= 7.04m^2$$

Total curved surface area for 24 pillar = 24 x curved surface of 1 pillar

$$= 24 \times 7.04 \text{ m}^2$$

$$= 168.96 \text{ m}^2$$

Rate of painting = ₹ 8 per m^2

Total cost = Total curved surface area x Rate of painting

$$= 168.96 \times ₹ 8$$

$$= ₹ 1,351.68$$

Hence, the total cost of painting the pillars is ₹ 1,351.68.

Question 12

The ratio between the curved surface area and the total surface area of a cylinder is 1 : 2. Find the volume of the cylinder, given its total surface area is 616 cm^2 .

Answer**Given:**

The ratio between the curved surface area and the total surface area of a cylinder is 1 : 2.

Total surface area of cylinder = 616 cm^2

$$\Rightarrow \frac{\text{Curved surface area of cylinder}}{\text{Total surface area of cylinder}} = \frac{1}{2}$$

$$\Rightarrow \frac{2\pi rh}{2\pi r(r+h)} = \frac{1}{2}$$

$$\Rightarrow \frac{\cancel{2\pi r}h}{\cancel{2\pi r}(r+h)} = \frac{1}{2}$$

$$\Rightarrow \frac{h}{r+h} = \frac{1}{2}$$

$$\Rightarrow 2 \times h = 1 \times (r+h)$$

$$\Rightarrow 2h = r+h$$

$$\Rightarrow 2h - h = r$$

$$\Rightarrow h = r$$

And,

$$\Rightarrow 2\pi r(r + h) = 616$$

$$\Rightarrow 2 \times \frac{22}{7} \times r(r + h) = 616$$

Using $r = h$

$$\Rightarrow 2 \times \frac{22}{7} \times r(r + r) = 616$$

$$\Rightarrow \frac{44}{7} \times r \times 2r = 616$$

$$\Rightarrow \frac{44}{7} \times 2r^2 = 616$$

$$\Rightarrow \frac{88}{7} \times r^2 = 616$$

$$\Rightarrow r^2 = \frac{7 \times 616}{88}$$

$$\Rightarrow r^2 = \frac{4,312}{88}$$

$$\Rightarrow r^2 = 49$$

$$\Rightarrow r = \sqrt{49}$$

$$\Rightarrow r = 7$$

And, volume of the cylinder = $\pi r^2 h$

$$= \frac{22}{7} \times 7^2 \times 7$$

$$= \frac{22}{\cancel{7}} \times 7^2 \times \cancel{7}$$

$$= 22 \times 49$$

$$= 1,078 \text{ cm}^3$$

Hence, the volume of the cylinder is $1,078 \text{ cm}^3$.

Question 13

The areas of three adjacent faces of a box are 120 cm^2 , 72 cm^2 and 60 cm^2 .

Find the volume of the box.

Answer

It is given that the areas of three adjacent faces of a box are 120 cm^2 , 72 cm^2 and 60 cm^2 .

Let l be the length, b be the breadth and h be the height of box.

$$\text{So, } l \times b = 120 \text{ cm}^2 \dots\dots\dots(1)$$

$$b \times h = 72 \text{ cm}^2 \dots\dots\dots(2)$$

$$h \times l = 60 \text{ cm}^2 \dots\dots\dots(3)$$

Multiplying equation (1), (2) and (3), we get

$$(l \times b) \times (b \times h) \times (h \times l) = 120 \times 72 \times 60 \text{ cm}^6$$

$$\Rightarrow l^2 \times b^2 \times h^2 = 518400 \text{ cm}^6$$

$$\Rightarrow l \times b \times h = \sqrt{518400} \text{ cm}^3$$

$$\Rightarrow l \times b \times h = 720 \text{ cm}^3$$

As we know that volume of box = $l \times b \times h$

$$\text{Volume of box} = 720 \text{ cm}^3$$

Hence, the volume of the box is 720 cm^3 .

Question 14

Eight identical cubes, each of edge 5 cm, are joined end to end. Find the total surface area and the volume of the resulting cuboid.

Answer

The length (l) of cuboid = $8 \times 5 \text{ cm} = 40 \text{ cm}$

The breadth (b) of cuboid = 5 cm

The height (h) of cuboid = 5 cm

As we know total surface area of the cuboid = $2(l \times b + b \times h + h \times l)$

$$= 2(40 \times 5 + 5 \times 5 + 5 \times 40) \text{ cm}^2$$

$$= 2(200 + 25 + 200) \text{ cm}^2$$

$$= 2 \times 425 \text{ cm}^2$$

$$= 850 \text{ cm}^2$$

As we know, volume of the cuboid = $l \times b \times h$

$$= 40 \times 5 \times 5 \text{ cm}^3$$

$$= 1,000 \text{ cm}^3$$

Hence, the total surface area of the cuboid is 850 cm^2 and the volume is $1,000 \text{ cm}^3$.

Question 15

A rectangular swimming pool 20 m long, 10 m wide and 2 m deep is to be tiled.
If each tile is 40 cm x 40 cm, find the number of tiles required.

Answer

Given:

Length of the swimming pool (l) = 20 m.

Breadth of the swimming pool (b) = 10 m.

Height of the swimming pool (h) = 2 m.

As we know that area of base of the swimming pool = $l \times b$

$$= 20 \times 10 \text{ m}^2$$

$$= 200 \text{ m}^2$$

The area of the four walls of the pool = $2(l + b)h$

$$= 2 (20 + 10) 2 \text{ m}^2$$

$$= 2 \times 30 \times 2 \text{ m}^2$$

$$= 120 \text{ m}^2$$

Total area of pool = Area of base + Area of four walls

$$\text{Total area of pool} = 200 + 120 \text{ m}^2$$

$$= 320 \text{ m}^2$$

Dimensions of each tile:

$$\text{Length} = 40 \text{ cm} = 0.4 \text{ m}$$

$$\text{Width} = 40 \text{ cm} = 0.4 \text{ m}$$

Area of one tile = Length x Width

$$= 0.4 \times 0.4 \text{ m}^2$$

$$= 0.16 \text{ m}^2$$

Let a be the number of tiles required.

$$a = \frac{\text{Total area of pool}}{\text{Area of one tile}}$$

$$\Rightarrow a = \frac{320}{0.16}$$

$$\Rightarrow a = \frac{32000}{16}$$

$$\Rightarrow a = 2,000$$

Hence, 2,000 tiles are required to tile the pool.

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